

Indian Institute of Engineering Science and Technology, Shibpur
Five Year Dual Degree (B. Tech.-M. Tech.) 3rd Semester Final Examination, 2017

Subject: Mathematics III (MA-301)

Full Marks: 70

Time: 3 hours

Use a separate answerscript for each half

1st Half (Full Marks - 40)

This half consists of questions worth 60 marks. Answer as much as you can or want to. The maximum you can score in this half is 40.

1. (a) A random variable X has probability density function (pdf) given by

$$f(x) = \begin{cases} kx^2 & 0 \leq x < 6 \\ k(12-x)^2 & 6 \leq x < 12 \\ 0 & \text{otherwise.} \end{cases}$$

Evaluate the constant k and find $P(X \leq 9 | X \geq 6)$.

- (b) Derive the normal equations for the simple linear regression model :

$$y_i = \alpha + \beta x_i + e_i, \quad i = 1, 2, \dots, n.$$

6 + 4 = 10

2. (a) State and prove Bayes' Theorem.

- (b) Suppose that $X \sim U(-\pi/2, \pi/2)$. Find the pdf of $Y = \tan X$.

6 + 4 = 10

3. (a) The probability that at least one of the events A and B occurs is $3/5$. If A and B occur simultaneously with probability $1/5$, then find $P(\bar{A}) + P(\bar{B})$, where $\bar{A}$, $\bar{B}$ are the complementary events of A , B respectively.

- (b) If the random variable X has a Binomial distribution with parameters n and p , find its mean and variance.

4 + 6 = 10

4. (a) Three groups of children contain respectively 3 girls and 1 boy, 2 girls and 2 boys and 1 girl and 3 boys. One child is selected at random from each group. Find the probability that the three selected children consist of 1 girl and 2 boys.

(b) State and prove Tchebycheff's inequality.

$$5 + 5 = 10$$

5. (a) The following is a random sample of size 10 from a uniform distribution on $(0, \theta)$, $\theta > 0$:
5.6, 3.9, 3.2, 4.8, 7.8, 6.1, 10.2, 8.4, 7.1, 9.8 . Find the Maximum Likelihood Estimator (MLE) of θ . Is the MLE an unbiased estimator for the parameter θ ? Justify your answer.

(b) State the Weak Law of Large Numbers.

$$8 + 2 = 10$$

6. (a) Prove that the correlation coefficient between two random variables always lies between -1 and 1.

(b) Define Mean Square Error (MSE) of an estimator. Establish the bias-variance decomposition formula for MSE.

$$5 + 5 = 10$$

SECOND HALF

ANSWER ANY THREE QUESTIONS IN THIS HALF

7. (a) Define Laplace transform of a function and find the Laplace transforms of the following functions:

i) $f(t) = te^{-t} \sin(3t)$

ii) $f(t) = (t+3)^2 e^t$

(b) If $L[f(t)] = F(s)$, then show that $L\left\{\int_0^t f(u)du\right\} = \frac{1}{s} F(s)$.

7+3=10

8. (a) State and prove convolution theorem for Laplace transform.

- (b) Find the inverse Laplace transforms of the following functions:

i) $F(s) = \frac{1}{(s^2+1)(s^2+9)}$

ii) $F(s) = \log\left(1 + \frac{a^2}{s^2}\right)$

4+6=10

9. (a) Solve the differential equation using Laplace transform

$$\frac{d^3 y}{dt^3} - 2\frac{d^2 y}{dt^2} + 5\frac{dy}{dt} = 0, \text{ with } y(0) = 0, y'(0) = 0, y''(0) = 1.$$

- (b) Show that the set of all feasible solutions of a linear programming problem is an convex set.

7+3=10

10. (a) Let $x_1 = 1, x_2 = 1, x_3 = 1$ and $x_4 = 0$ be a feasible solution(F.S.) of the system

$$x_1 + 2x_2 + 4x_3 + x_4 = 7$$

$$2x_1 - x_2 + 3x_3 - 2x_4 = 4.$$

Reduce the above F.S. into two different B.F.S.

- (b) Solve the following L.P.P. graphically:

Minimize $Z = x_1 + x_2$

subject to $5x_1 + 9x_2 \leq 45,$

$$x_1 + x_2 \geq 2,$$

$$x_1 \leq 4,$$

$$x_1, x_2 \geq 0.$$

6+4=10

11. Using Charne's Big M method, solve the following L.P.P. :

$$\text{Minimize } Z = 4x_1 + x_2$$

subject to

$$3x_1 + x_2 = 3,$$

$$4x_1 + 3x_2 \geq 6,$$

$$x_1 + 2x_2 \leq 4,$$

$$x_1, x_2 \geq 0.$$

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OR

Apply two phase method to show that the following L.P.P. has unbounded solution:

$$\text{Maximize } Z = 2x_1 - x_2 + 2x_3$$

subject to

$$x_1 + x_2 - 3x_3 \leq 8,$$

$$4x_1 - x_2 + x_3 \geq 2,$$

$$2x_1 + 3x_2 - x_3 \geq 4,$$

$$x_1, x_2, x_3 \geq 0.$$

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